GENETICS OF LIFE

PYQ • 2021 - 2026

1. 2026 **1 MARK**

Which option correctly lists all three components of a single DNA nucleotide ?

(a) Phosphate, two nitrogen bases and a deoxyribose sugar

(b) Deoxyribose sugar, a nitrogen base and a phosphate

(c) Deoxyribose sugar, two phosphates and a nitrogen base

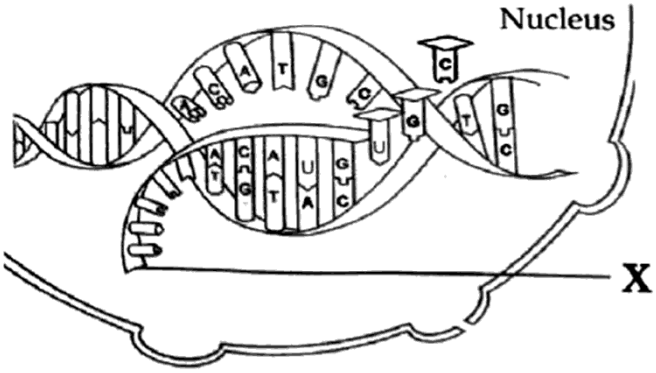
(d) Deoxyribose sugar, phosphate and an amino acid

2. 2026 **2 MARKS**

Observe the illustration and answer the following questions.

(a) Identify the molecule labelled as 'X'.

(b) Mention its role in protein synthesis.



3. 2026 **3 MARKS**

(A) A tall round seeded plant (TTRR) is crossed with a dwarf wrinkled seeded plant (ttrr).

(a) Write the genotype and phenotype of F_1 plants.

(b) If F_1 plants are self-pollinated, write the possible phenotypes in F_2.

OR

(B) By giving suitable examples based on the hints, explain various situations which differ from Gregor Mendel's hypothesis.

Hints : (a) Incomplete dominance (b) Co-dominance (c) Polygenic inheritance

4. 2026 **3 MARKS**

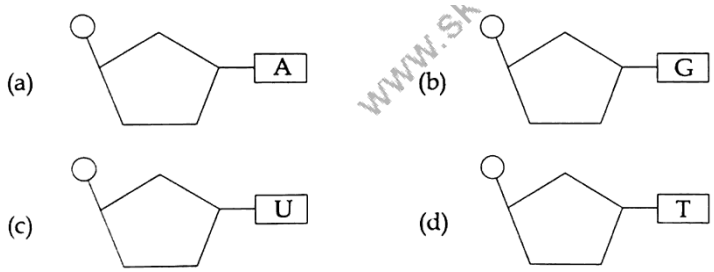
A patient shows frequent fatigue and weakness. Blood examination reveals that some red blood cells are sickle-shaped.

(a) Identify the reason for this disease.

(b) In what other ways does this genetic defect affect the body ?

5. 2025 **1 MARK**

Which of the following nucleotides is found only in RNA ?



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6. 2025 **2 MARKS**

Analyse each of the given situations and explain how variations are formed in the offspring.

(a) During the initial phase of meiosis, chromosomes pair.

(b) Parents' chromosomes reach their offspring through gametes.

7. 2025 **3 MARKS**

Answer the following questions related to an experiment in which a tall pea plant (TT) is crossed with a dwarf pea plant (tt).

(a) Write the genetic structure and character of the first generation.

(b) Illustrate the experiment using symbols in which the second generation is produced.

8. 2024 **1 MARK**

Hybridisation of two contrasting traits of the character seed colour is illustrated below.

Yellow seed (YY) X Green seed (yy)

Which will be the recessive trait of the F_1 generation ?

9. 2024 **2 MARKS**

Analyse the given condition and answer the questions :

'Red blood cells bend like sickle'

(a) Identify the disease.

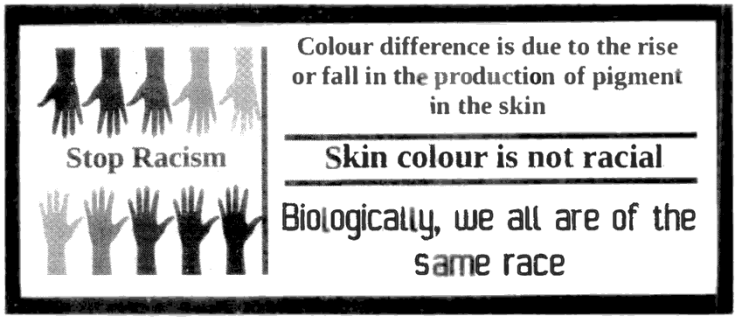
(b) In this condition, other blood cells do not undergo the same structural change as red blood cells. Why ?

10. 2024 **2 MARKS**

Observe the poster against colour discrimination and answer the questions.

(a) Name the pigment mentioned in the poster.

(b) What is the reason for the rise and fall in the production of this pigment ?



11. 2024 **3 MARKS**

Opinions of three students on protein synthesis are given below. Analyse them and answer the question.

Student 1 : Protein synthesis begins in the cytoplasm and ends in the nucleus.

Student 2 : Protein synthesis begins in the nucleus and ends in the cytoplasm.

Student 3 : The entire process of protein synthesis takes place in the nucleus.

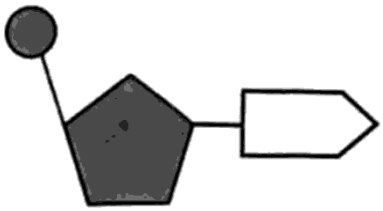
Whose opinion is correct ? Explain why ?

12. 2023 **1 MARK**

The genetic make up of human female is 44 + XX and that of male is 44 + XY. What do the XX and XY indicate ?

13. 2023 **1 MARK**

What does the illustration indicate ?

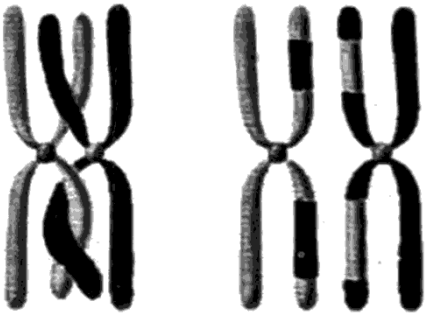


14. 2023 **2 MARKS**

Analyse the given illustration and answer the questions.

(a) Which process does this illustration indicate ?

(b) How does this process cause variation ?



15. 2023 **3 MARKS**

Stages in the process of protein synthesis are given below. Arrange them in sequential order.

- Protein is synthesized.
- mRNA reaches ribosome.
- mRNA reaches outside the nucleus.
- Amino acids are added based on information in mRNA.
- tRNA brings different kinds of amino acids to ribosome.
- mRNA forms from DNA.

16. 2022 **1 MARK**

Find out the genetic constitution of the sperm.

(a) 44 + X (b) 22 + XX (c) 44 + XX (d) 22 + Y

17. 2022 **2 MARKS**

Analyse the given genetic makeup and answer the questions.

44 + XX

(a) Give the number of somatic chromosomes.

(b) Is the individual with this genetic makeup male or female ? Why ?

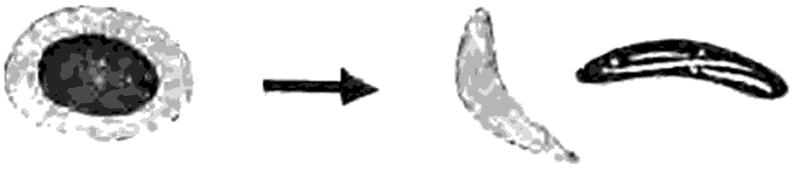
18. 2022 **4 MARKS**

Change in the shape of red blood cells in a genetic disease is given below. Analyse it and answer the questions.

(a) Name the disease indicated.

(b) Why do the shape of red blood cells change ?

(c) How does the deformity of red blood cells affect the body ?



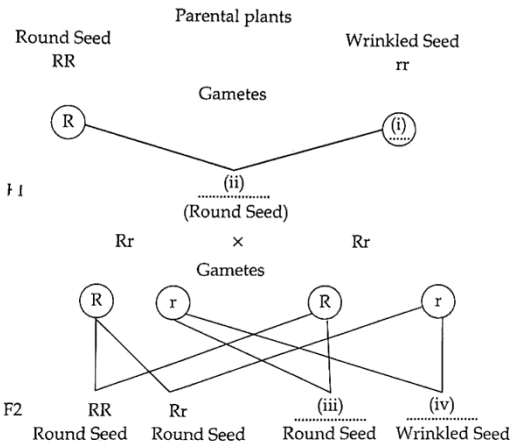
19. 2022 **4 MARKS**

Observe the illustration given below and answer the questions. (Parental plants : Round seed RR X Wrinkled seed rr)

(a) Fill (i), (ii), (iii), (iv).

(b) Which is the dominant character ?

(c) Why does the recessive trait in the first generation appear in the second generation ?



20. 2021 **1 MARK**

Complete the illustration according to the Model.

Model : RNA – Ribose sugar

DNA – _____

21. 2021 **2 MARKS**

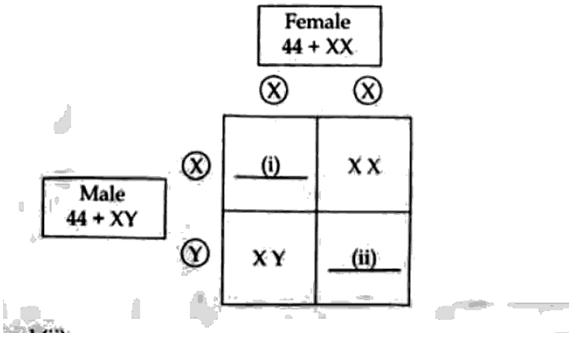
What is the role of mRNA and tRNA in protein synthesis ?

22. 2021 **2 MARKS**

Analyse the illustration and answer the questions.

(a) Fill up (i) and (ii).

(b) 'The chromosome of the father determines the sex of the child'. Explain.



23. 2021 **3 MARKS**

Observe the illustration and answer the questions.

(a) Identify the illustration.

(b) What do A and B indicate ?

(c) Name the type of 'B' found only in DNA molecule.

